

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Nutrient pollution, sewage treatment & acid mine drainage - Set 1

Mark scheme

AQA spec 3.4.3.2.9-3.4.3.2.10; Pollution; textbook pp. 266-275

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Dissolved oxygen fluctuates daily / 24-hour cycle: highest in the afternoon (about 13-13.5 mg dm⁻³ at 16 and 40 h) and lowest at night / early morning (2.5-3.5 mg dm⁻³ at 4 and 28 h) • The night-time minimum is lower on the second night (2.5 vs 3.5)	2
01.2	• Nitrates (from fertiliser runoff) and phosphates (from sewage effluent) cause eutrophication: rapid growth of algae / cyanobacteria (algal bloom) • During the day photosynthesis by the dense algae releases oxygen so concentrations rise (above the normal saturation level) • At night there is no photosynthesis but respiration by algae / animals / bacteria continues, using oxygen • Aerobic bacterial decomposition of dead algae / shaded macrophytes uses oxygen; warm summer water holds less oxygen and decay is rapid	4
01.3	• At night no photosynthesis replaces the oxygen used in respiration / decomposition so concentrations fall below the minimum fish can tolerate • At the end of summer large amounts of dead vegetation decay rapidly in warm water causing severe deoxygenation	2
01.4	<i>Any 2 from:</i> • Cyanobacteria release toxins harmful to livestock, pets and people • Floating algae shade submerged / rooted macrophytes which die, disrupting food webs • Algae are an unreliable food source (die off rapidly) so food webs are poor • Blocked filters / unpleasant taste and smell in water supplies	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 266-268; AQA spec 3.4.3.2.9.	

Question	Marking guidance	Marks
02.1	• Bacteria in the gut convert nitrates to nitrites • Nitrites react with haemoglobin so it cannot carry oxygen • Infants are most at risk because they have more of the gut bacteria and their haemoglobin reacts faster; usually from shallow wells receiving arable drainage	3
02.2	• Nitrites may be converted in the gut to nitrosamines which are carcinogenic • But most research has been on rats / the evidence in humans is not conclusive; nitrites are also ingested in processed food so the source cannot be isolated	2
02.3	<i>Any 3 from:</i> • Reduce use of nitrate fertilisers / match application to crop need • Avoid ploughing / applying fertiliser at times of heavy rain (autumn / winter) • Use low-solubility / slow-release fertilisers rather than soluble rapid-release ones • Store manure / slurry where it cannot leach or wash into rivers • Grow crops with lower nitrogen requirements (eg triticale instead of wheat) • Grow legumes with nitrogen-fixing bacteria instead of relying on artificial fertiliser	3
02.4	• Iron(III) sulfate is added to effluent: phosphates are precipitated as insoluble iron phosphate (a fine sediment that can be used as fertiliser) • Lakes: dredging phosphate-rich sediments / adding iron(III) sulfate to reduce phosphate solubility (eg the Norfolk Broads)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 267-268; AQA spec 3.4.3.2.9.	
03.1	• Screens (metal grills) trap floating / suspended non-biodegradable items (paper, plastic) which are landfilled or incinerated • Grit traps: the channel widens so flow slows and stones / road grit sink out because there is not enough kinetic energy to carry them	2
03.2	• The effluent stands in large tanks so the faecal / organic solids sink and are removed as sludge • This removes over 95% of the organic matter from the fluid so the risk of deoxygenation of the river is greatly reduced	2
03.3	• Aerobic bacteria digest / break down the remaining dissolved and suspended organic matter • Air (or pure oxygen) is mixed in by paddle wheels or air stones so oxygen shortage does not limit the aerobic process • The suspended bacteria collected in the secondary sedimentation tank are the 'activated sludge'; returning them maintains a large bacterial population so digestion is rapid	3
03.4	• Effluent from secondary treatment may still contain dissolved inorganic nutrients (phosphates, nitrates) that cause eutrophication, and some bacteria • Needed where the discharge site is ecologically sensitive or used by many people (bathing water, drinking water abstraction) • Phosphate stripping with iron(III) sulfate; micro-strainers (very fine sieves) then UV light or chlorine to kill remaining bacteria	3

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 269-272; AQA spec 3.4.3.2.9.	
04.1	<i>Any 4 from:</i> • Sludge from primary sedimentation is digested by anaerobic microbes in a warm tank for about four weeks • Kills most pathogens • Reduces odour and volume of the sludge • Produces biogas (mainly methane with some CO₂) used as fuel to heat the digester / generate electricity • Digested sludge has a better C:N ratio and can be used as agricultural fertiliser	4
04.2	• Heavy metals from urban wastes / road runoff may be present and absorbed by crops - reduced by adding lime to lower metal solubility • Remaining pathogens - destroyed by heat treatment of the sludge before use • Timing: spreading in the growing season covers crops; in winter vehicles compact wet soils - apply at suitable times / when soils are dry • Odour / public objection - digestion reduces odour	4
04.3	• Foul waste is a small volume needing full treatment so smaller, cheaper treatment works can be built; surface water needs little treatment • The treatment works is not affected by weather-related fluctuations in flow (storms) so storm overflows of untreated sewage are avoided	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 270-273; AQA spec 3.4.3.2.9.	
05.1	• Spoil contains residual sulfide ores (eg iron pyrites, zinc sulfide) • These react with oxygen in the air spaces between the waste particles (oxidation) • Producing metal oxides and oxides of sulfur which dissolve in drainage water to form sulfuric acid	3
05.2	• The low pH (3.4) directly damages exposed cells / denatures proteins and dissolves calcium-based exoskeletons • Low pH increases the solubility of zinc (0.02 to 6.5 mg dm⁻³) which is toxic (inhibits enzymes) • Only a few tolerant taxa survive; the sensitive mayfly larvae (indicator of clean water) are eliminated; dissolved oxygen is not the limiting factor here	3
05.3	• Dilution by tributaries / more water • Neutralisation by alkaline rocks (limestone) in the stream bed; zinc precipitates / adsorbs onto sediments as pH rises	2
05.4	• Collect the drainage water and pass it through crushed limestone / add lime to neutralise the acid • Higher pH also reduces the solubility of toxic metals so they precipitate and can be removed	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 273-275; AQA spec 3.4.3.2.9-3.4.3.2.10.	

Question	Marking guidance	Marks
06.1	• Many different organic substances may be present, each with a different potential to cause deoxygenation, so it is difficult to measure or forecast them individually • BOD measures the effect (amount of oxygen used by microbes decomposing the organic matter) under standardised conditions (eg 5 days, 20 °C, dark) • It gives a single measure directly related to the risk of deoxygenation	3
06.2	• E. coli is a common gut bacterium so its presence confirms faecal contamination and its number indicates how serious the contamination is • Pathogens are not always present so their absence would not prove the water is uncontaminated	2
06.3	<i>Any 3 from:</i> • Different species have different sensitivities to pollution • Easy to identify • Easy to find / normally present / usually common • Widely distributed	3
06.4	• Advantage: assesses the effect of pollution over a period of time (recent pollution events are detected even if the pollutant has gone); no expensive equipment needed • Disadvantage: identification needs skill; does not identify which pollutant is present so further chemical tests are required; species may be absent for other reasons	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 273-275; AQA spec 3.4.3.2.9.	

Question	Marking guidance	Marks
07.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Organic nutrients (sewage, manure, silage effluent, paper mill / tannery / food processing effluent) are food for aerobic microbes whose respiration deoxygenates the water, killing fish and invertebrates; bacterial growths cover the bed, blocking light • Pathogens in sewage (cholera, typhoid, dysentery) spread via contaminated water; decay of organic matter releases inorganic nutrients • Inorganic nutrients: nitrates (fertiliser / manure runoff, leaching) and phosphates (sewage effluent, detergents, eroded soil) cause cultural eutrophication: algal blooms, shading of macrophytes, cyanobacterial toxins, night-time and end-of-summer deoxygenation • Human health: nitrites and methaemoglobinaemia; possible carcinogen via nitrosamines • Control of organic wastes: sewage treatment (pre-treatment, primary sedimentation removing >95% organic matter, aeration / activated sludge or trickling filters, anaerobic digestion of sludge with biogas) • Control of nutrients: tertiary phosphate stripping with iron(III) sulfate, denitrification in filter beds, micro-strainers / UV; Nitrate Vulnerable Zones, fertiliser timing / amount, low-solubility fertilisers, legumes, manure management; dredging sediments • Evaluation: sewage treatment has virtually eliminated deoxygenation and waterborne disease in developed countries but storm overflows and combined sewers still discharge untreated sewage; tertiary treatment is expensive so only used at sensitive sites; diffuse agricultural nitrate is hard to control and aquifers respond slowly; phosphate legacy in sediments • Judgement on point-source (sewage) control being more successful than diffuse-source (agriculture) control	9
07.2	• Algae / cyanobacteria absorb nutrients directly from the water (not the sediments) so they use dissolved nutrients more easily and grow rapidly	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 266-272; AQA spec 3.4.3.2.9.	